MASTER'S DEGREE IN PHARMACEUTICAL BIOTECHNOLOGY

From Genome to Drug – Exploring Bioinformatics & Omics Sciences

Lesson I – Part II
Sequence Alignment, BLAST & Practical Tools

Bioinformatics Databases and Alignment Tools

Estevão Barcelos, PostDoc UNIPG

estevaocarlos.silvabarcelos@collaboratori.unipg.it

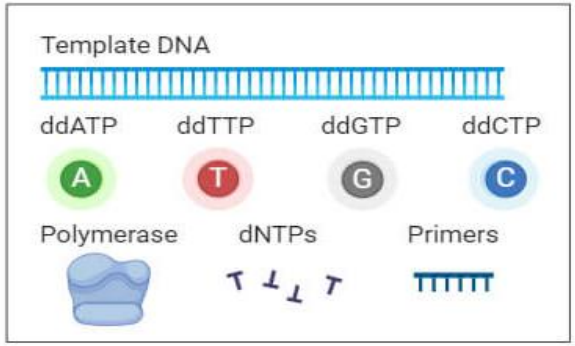
Learning Objectives — Lesson II

- Sequence alignment is and why it matters in biology
- Use BLAST to identify unknown sequences and interpret E-value, % identity, coverage
- Distinguish global alignment (Needleman-Wunsch) from local alignment (Smith-Waterman)
- Understand mutations: missense, nonsense, silent and frameshift — and their protein impact

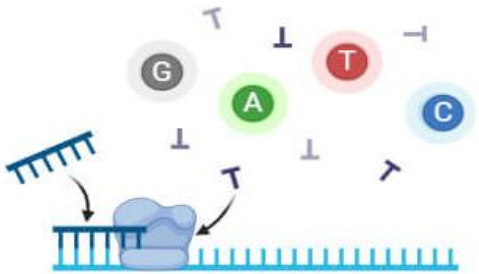


Sanger Sequencing

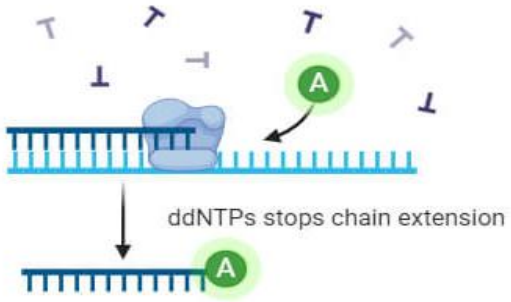
Reagents



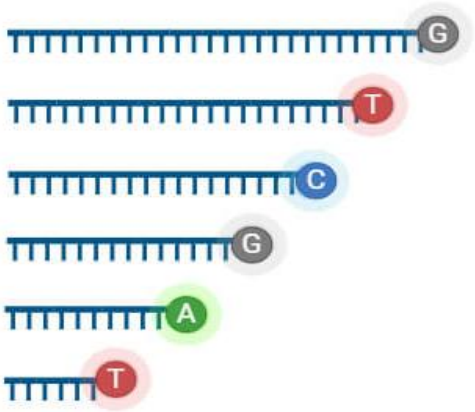
① Primer annealing and chain extension



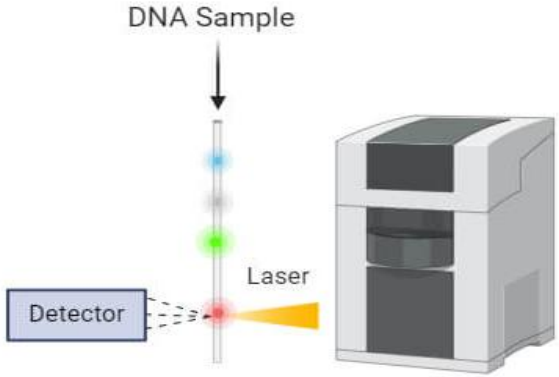
② ddNTP binding and chain termination



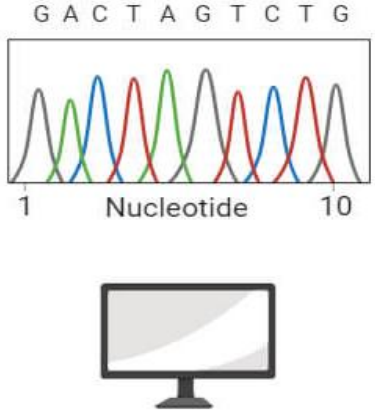
③ Fluorescently labelled DNA sample



④ Capillary gel electrophoresis and fluorescence detection



⑤ Sequence analysis and reconstruction

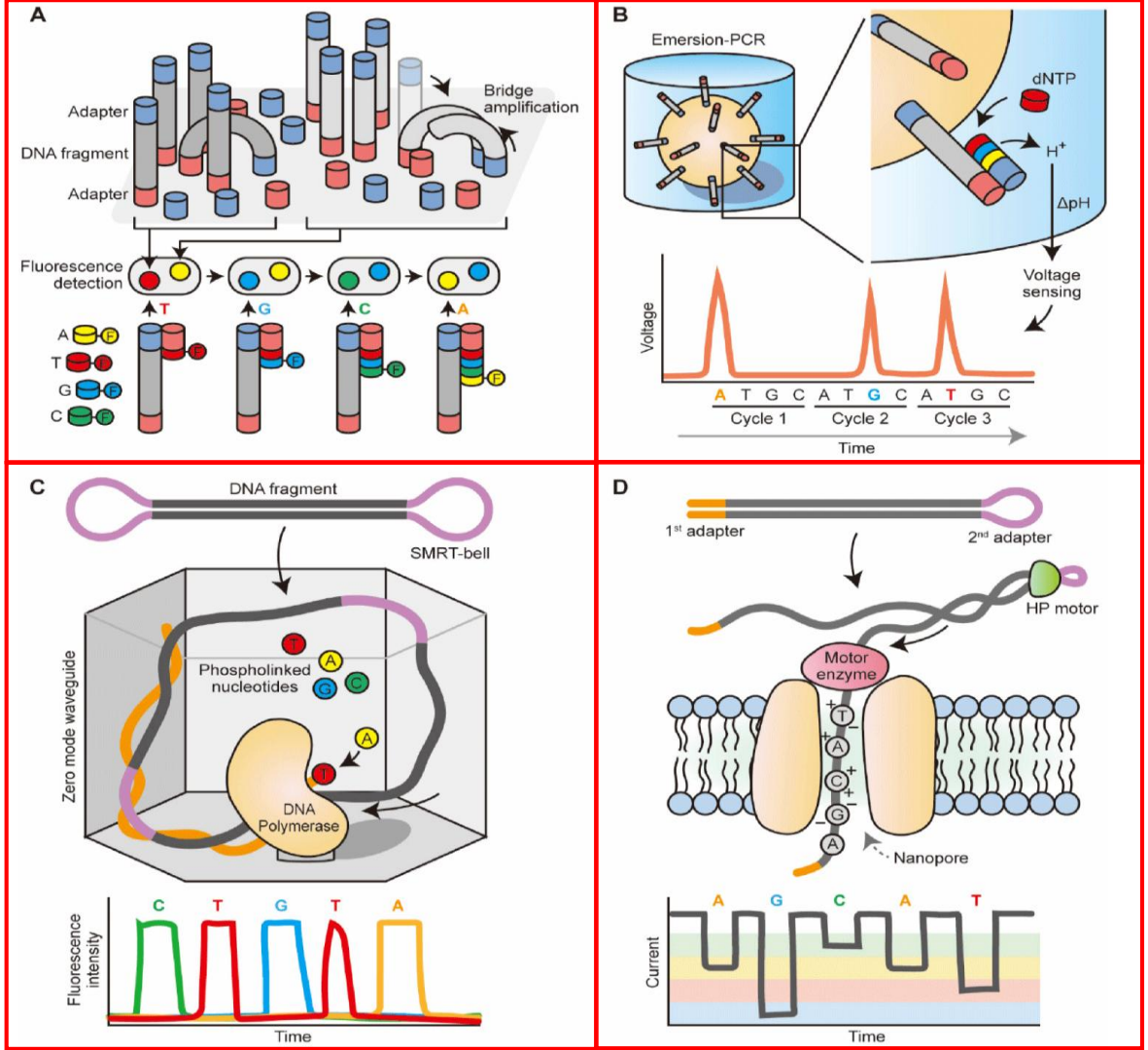


illumina®

ion torrent
by Thermo Fisher Scientific

PacBio

Oxford Nanopore Technologies





Which Sequencing to Choose?

Apply your sequencing knowledge to real clinical and research scenarios. Choose the appropriate technique and justify your choice.

Which Sequencing Technique Would You Use?

CASE 1

A clinician wants to confirm a single known mutation in the *BRCA1* gene in a patient sample. The lab needs a result **by end of day**, with **very high base-level accuracy**. Only one specific position (out of 3 billion) needs to be checked.

→ Which Sequencing?

CASE 2

A research group wants to sequence **500 human whole genomes** from a population cohort to **identify common and rare SNPs associated with autoimmune disease.** Budget matters — cost per genome must be **as low as possible**.

→ Which Sequencing?

CASE 3

A hospital in a **remote area in West Africa** receives patients during an **outbreak of an unknown pathogen**. A field scientist has **a laptop and a USB-powered device**. She needs to sequence the pathogen genome **on-site within hours to identify the virus** and inform containment decisions.

→ Which Sequencing?

Sanger sequencing

Gold standard for confirming a single known variant.

Illumina

Illumina dominates population-scale WGS. Its cost fell from $3 billion (HGP) to under $200, and its throughput.

Nanopore (MiniON)

The MinION was designed exactly for this scenario. It ran on a laptop in a tent during the 2015 West Africa Ebola outbreak, providing same-day sequencing data that informed containment responses.

Which Sequencing Technique Would You Use?

CASE 4

A genomics centre wants to finally close the remaining **8% gaps in the human reference genome** — regions **filled with long, complex tandem repeats that Illumina cannot assemble**. They need reads long enough to span entire repeat arrays.

→ **Which Sequencing?**

PacBio (HiFi) + Oxford Nanopore (combined)

*The **Telomere-to-Telomere (T2T) Consortium** completed the first truly gapless human genome in 2022 using exactly this combination. **PacBio HiFi provided high-accuracy long reads; Nanopore provided ultra-long reads to span the most complex regions.** Neither alone was sufficient.*

CASE 5

A clinical oncology laboratory receives a tumour biopsy. The oncologist needs a rapid targeted panel — **checking 50 cancer-relevant genes (including CTLA4, TP53, KRAS, BRAF) for actionable mutations**. Results are needed within **the same working day** to inform **treatment**.

→ **Which Sequencing?**

Ion Torrent

Ion Torrent's speed — runs complete in 2–4 hours — and its semiconductor chip design make it the standard choice for rapid clinical targeted panels. Its proton-detection chemistry eliminates cameras and lasers, making the instrument compact enough to sit on a bench in a hospital pathology lab.

CASE 6

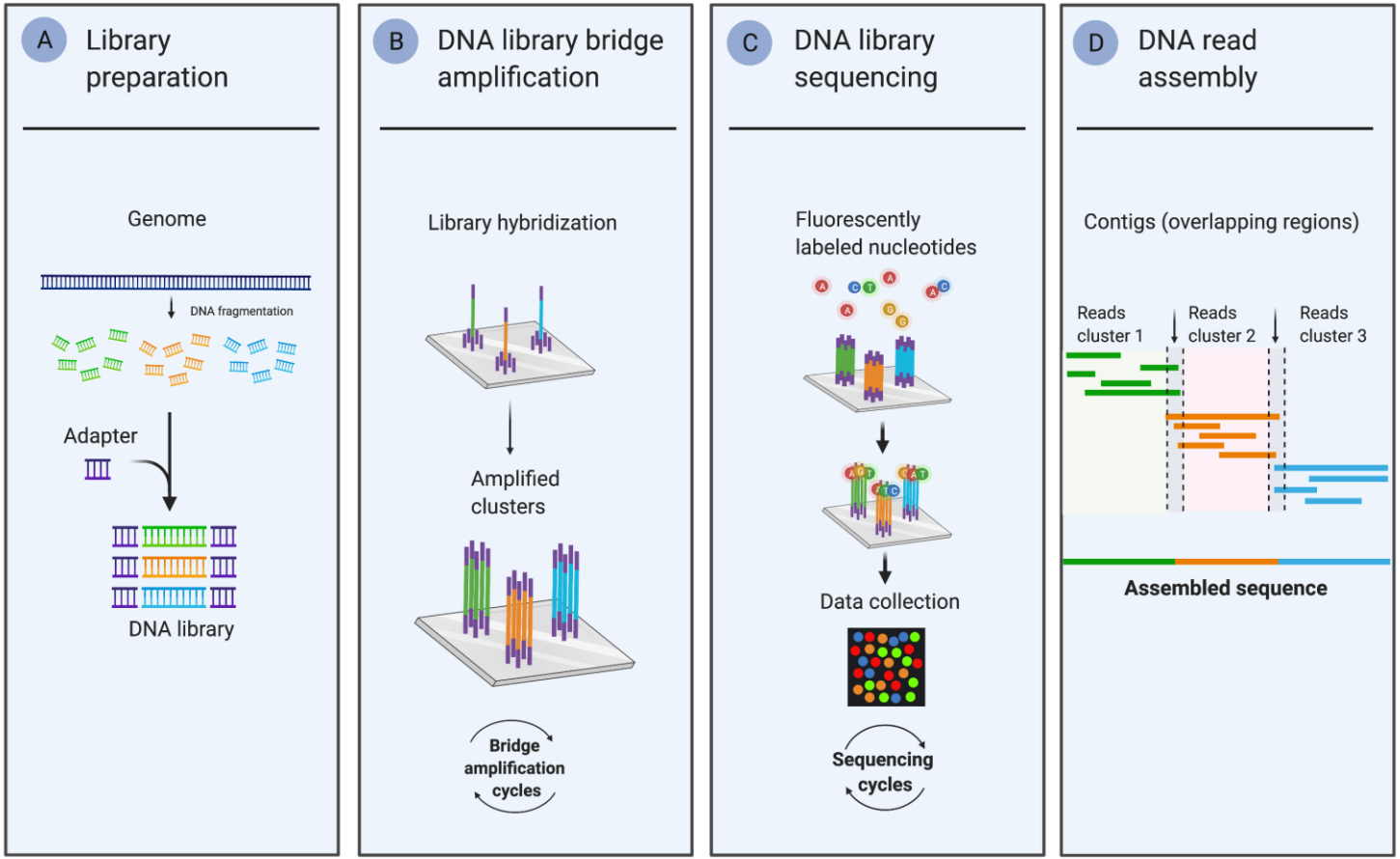
A biotechnology company has engineered a bacterial clone to express a recombinant protein.

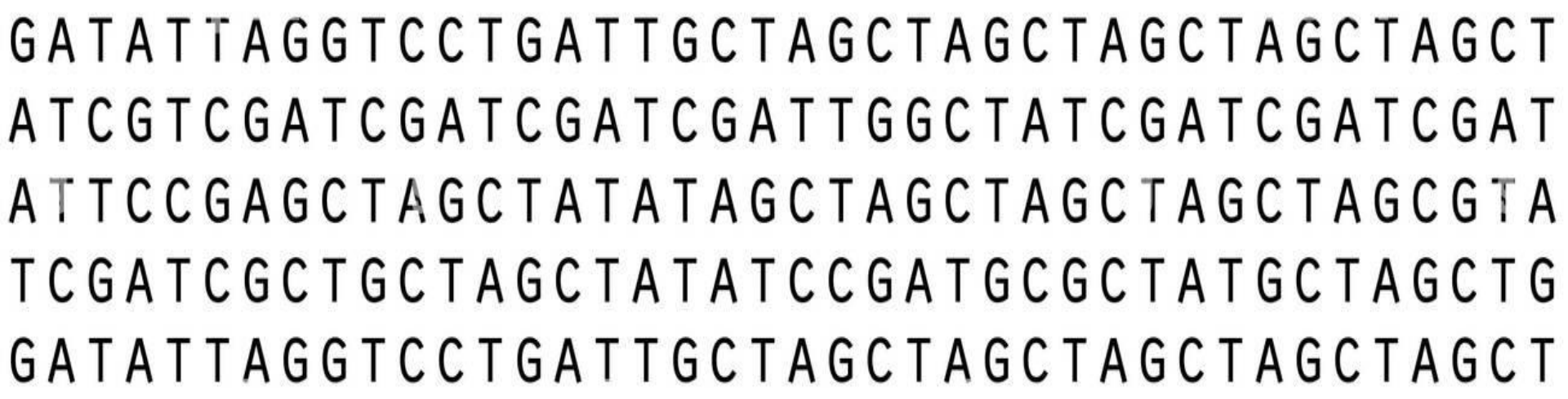
Before scaling up production, they need to verify that **the insert sequence** is exactly correct — **no point mutations, no frameshifts. The insert is 800 bp.** Which technology?

→ **Which Sequencing?**

Sanger sequencing

For an 800 bp insert, Sanger sequencing provides high accuracy for detecting single-nucleotide (point) mutations, insertions, and deletions that can cause frameshifts.-

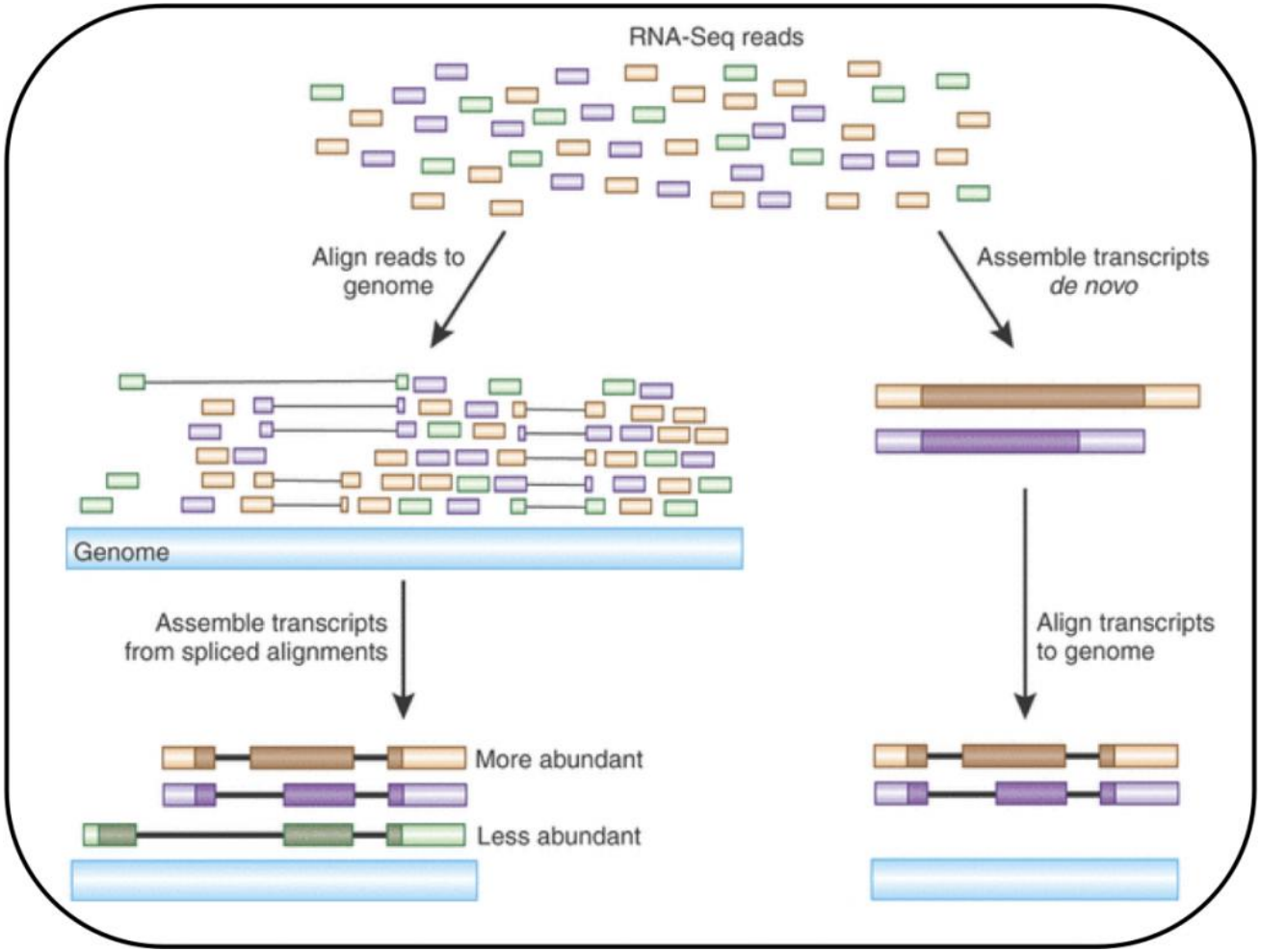

A
Library preparation
Genome
DNA fragmentation
Adapter
DNA library
B
DNA library bridge amplification
Library hybridization
Amplified clusters
Bridge amplification cycles
C
DNA library sequencing
Fluorescently labeled nucleotides
Data collection
Sequencing cycles
D
DNA read assembly
Contigs (overlapping regions)
Reads cluster 1
Reads cluster 2
Reads cluster 3
Assembled sequence

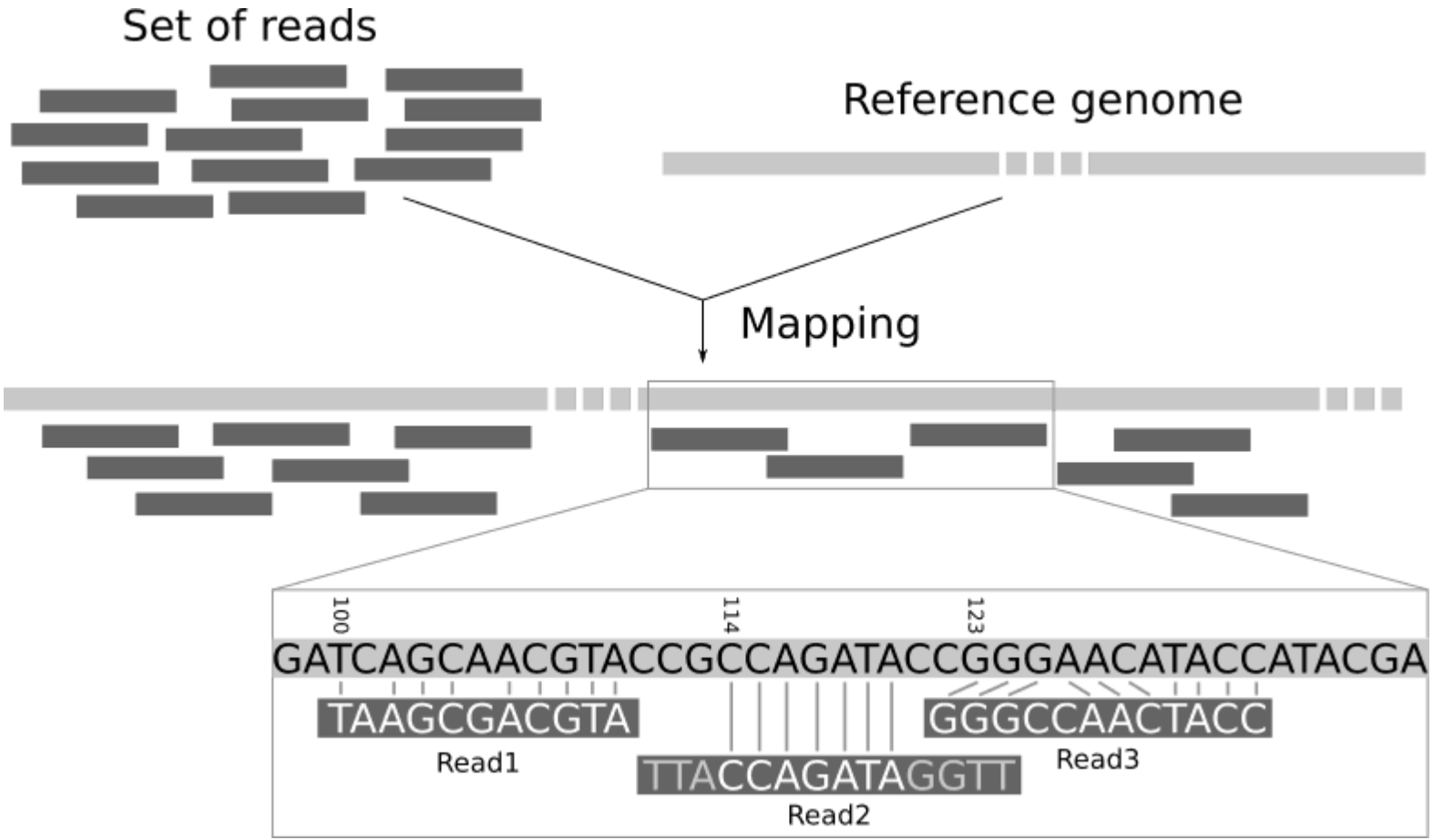


“Okay, we’ve sequenced the DNA. Now we have billions of letters.

So what?”

Sequence Assembling/ Mapping




Set of reads
Reference genome
Mapping
100
114
123
GATCAGCAACGTACCGCCAGATACCGGGAACATACCATACGA
TAAGCGACGTA
Read1
TTACCAGATAGGTT
Read2
GGGCCAACTACC
Read3



Sequence Alignment

“Sequencing reads one sequence. Alignment compares two.”

What Is Sequence Alignment?

*Arranging two or more sequences to identify regions of similarity — **revealing functional, structural, or evolutionary relationships.***

Functional

Similar sequences often share biological function. CTLA-4 and CD28 share the Ig V-set domain — both bind CD80/CD86, but with opposite effects on T cell activation.

Structural

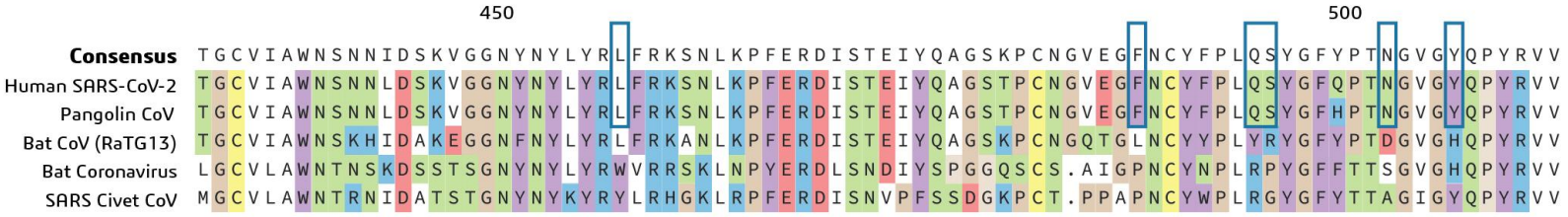
Conserved sequence usually means conserved 3D shape. Even proteins with <30% identity can fold identically if key structural residues are conserved.

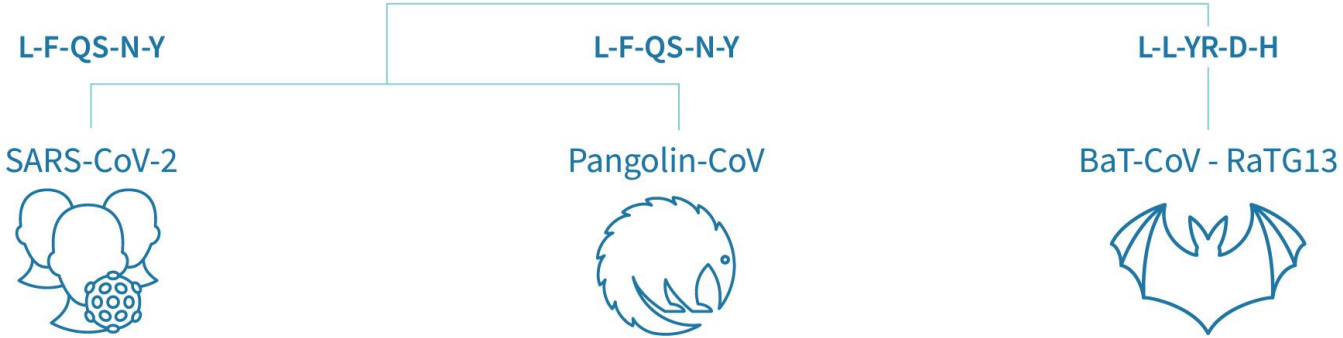
Evolutionary

Similarity reflects common ancestry. Human and mouse CTLA-4 share ~76% identity — both inherited it from a common mammalian ancestor ~80 million years ago.

Key principle: sequences conserved across millions of years of evolution are functionally essential — mutating them is usually lethal.

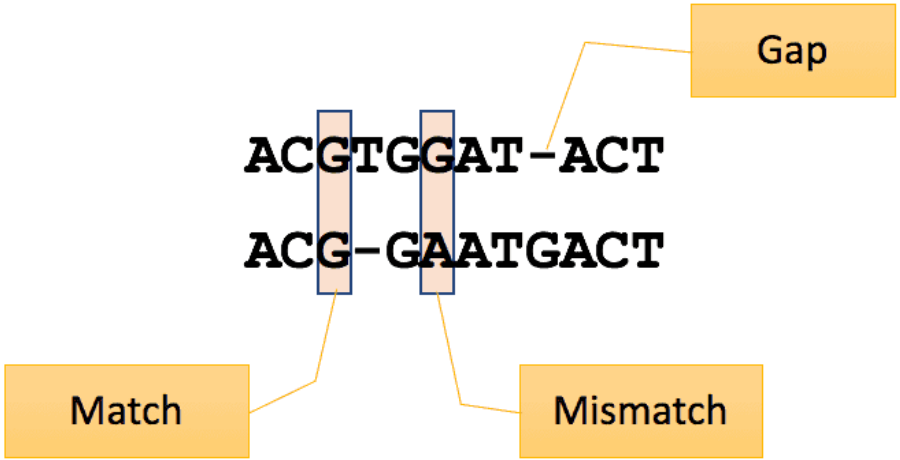
Comparison between sequences





SARS-CoV-1 and **MERS-CoV** share approximately **79.6%** and **50%** whole-genome nucleotide identity with **SARS-CoV-2**, respectively. In comparison, **bat coronavirus RaTG13** shares **96.2% identity**, while **pangolin coronaviruses** share approximately **85–92%**. Notably, some **pangolin-CoVs** exhibit up to **97.4% amino acid similarity to SARS-CoV-2** in **the spike receptor-binding domain (RBD)**, exceeding that of RaTG13 (Zhou et al., 2020; Lam et al., 2020; Xiao et al., 2020).

“We have two DNA sequences. They are clearly related — they look similar. But they are not identical.
Our job is to compare them systematically, position by position, and classify every column into one of three categories.”

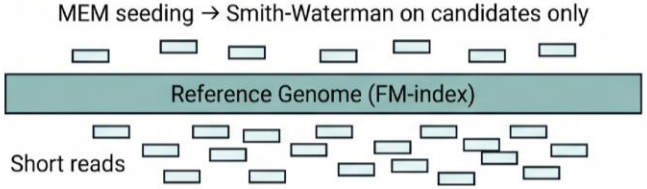


$$Similarity(\%) = \frac{Match}{Match + Mismatch} \times 100$$

$$Above\ case: \ \frac{1}{9+1} \times 100 = 90\%$$

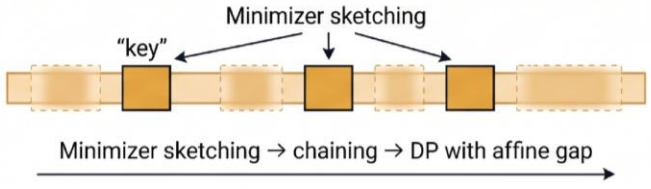
Sequence Alignment - Dynamic Programming

BWA-MEM — Short Reads (Illumina)



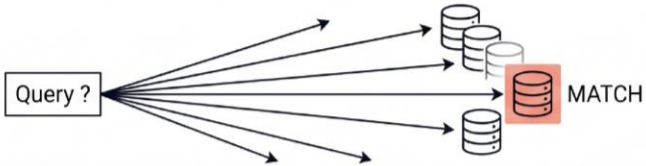
Confetti from a known book — find the exact page

Minimap2 — Long Reads (Nanopore / PacBio)



Long torn strip — use the still-readable words to find your place

BLAST — Database Search

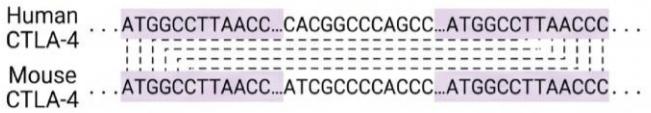


k-mer seeds → ungapped extension → Smith-Waterman

A page with no cover — search the whole library

EMBOSS Needle — Global Pairwise

Needleman-Wunsch — full DP matrix — mathematically optimal



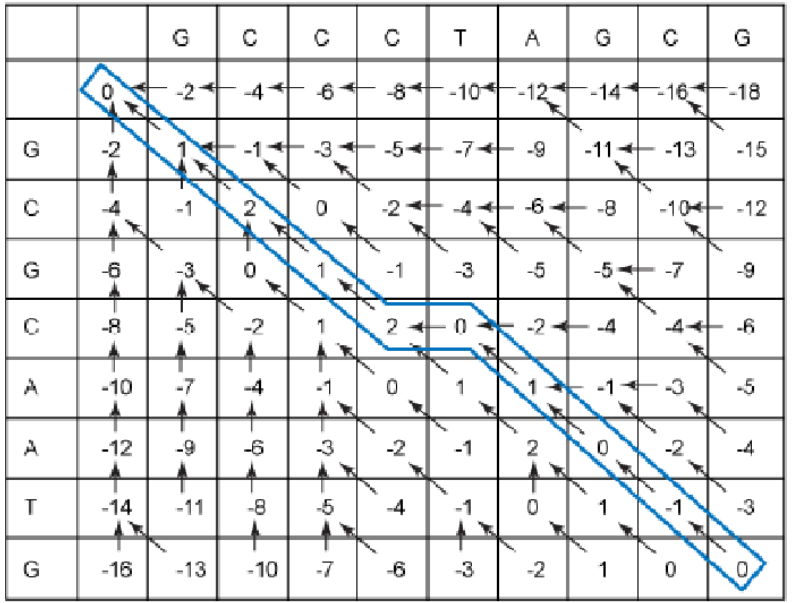
Two books open side by side — compare line by line

All use dynamic programming — what changes is WHERE and WHEN it is applied

Dynamic Programming — The Logic Behind Alignment

❗ *The idea: break a complex alignment problem into many small, overlapping sub-problems. Solve each small problem once and build up to the full solution.*

- The **Needleman–Wunsch algorithm** performs a global alignment of sequences.
- It was one of the first applications of dynamic programming to compare biological sequences.
- The algorithm was developed by Saul B. Needleman and Christian D. Wunsch and published in 1970.

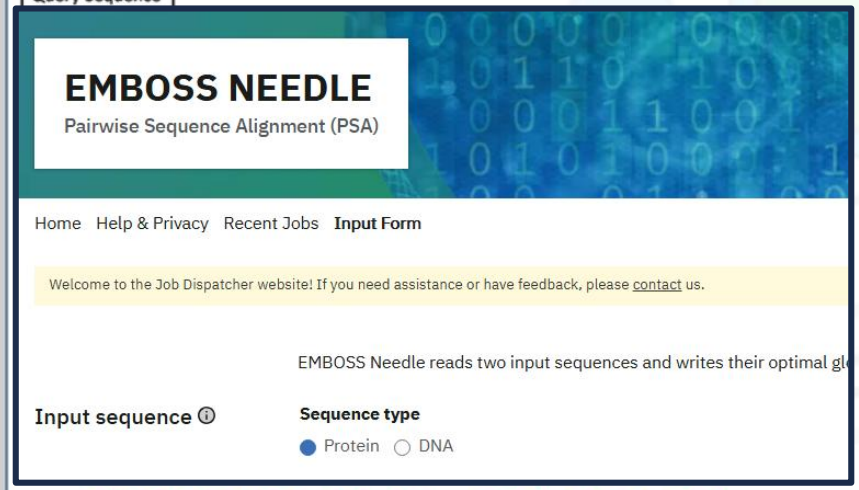


		G	C	C	C	T	A	G	C	G
	0	-2	-4	-6	-8	-10	-12	-14	-16	-18
G	-2	1	-1	-3	-5	-7	-9	-11	-13	-15
C	-4	-1	2	0	-2	-4	-6	-8	-10	-12
G	-6	-3	0	1	-1	-3	-5	-5	-7	-9
C	-8	-5	-2	1	2	0	-2	-4	-4	-6
A	-10	-7	-4	-1	0	1	1	-1	-3	-5
A	-12	-9	-6	-3	-2	-1	2	0	-2	-4
T	-14	-11	-8	-5	-4	-1	0	1	-1	-3
G	-16	-13	-10	-7	-6	-3	-2	1	0	0

Alignment

Global Alignment

Target Sequence
5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'
||||||||||| ||||||| |||||||||||||| |||||||
5' ACTACTAGATT----ACGGATC--GTACTTTAGAGGCTAGCAACCA 3'
Query Sequence

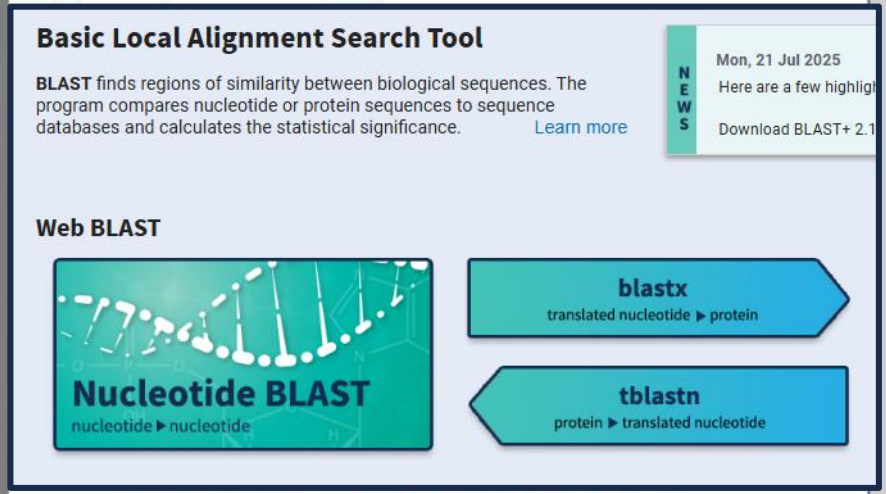


NEEDLEMAN - WUNSCH

GLOBAL VS. LOCAL SEQUENCE ALIGNMENT

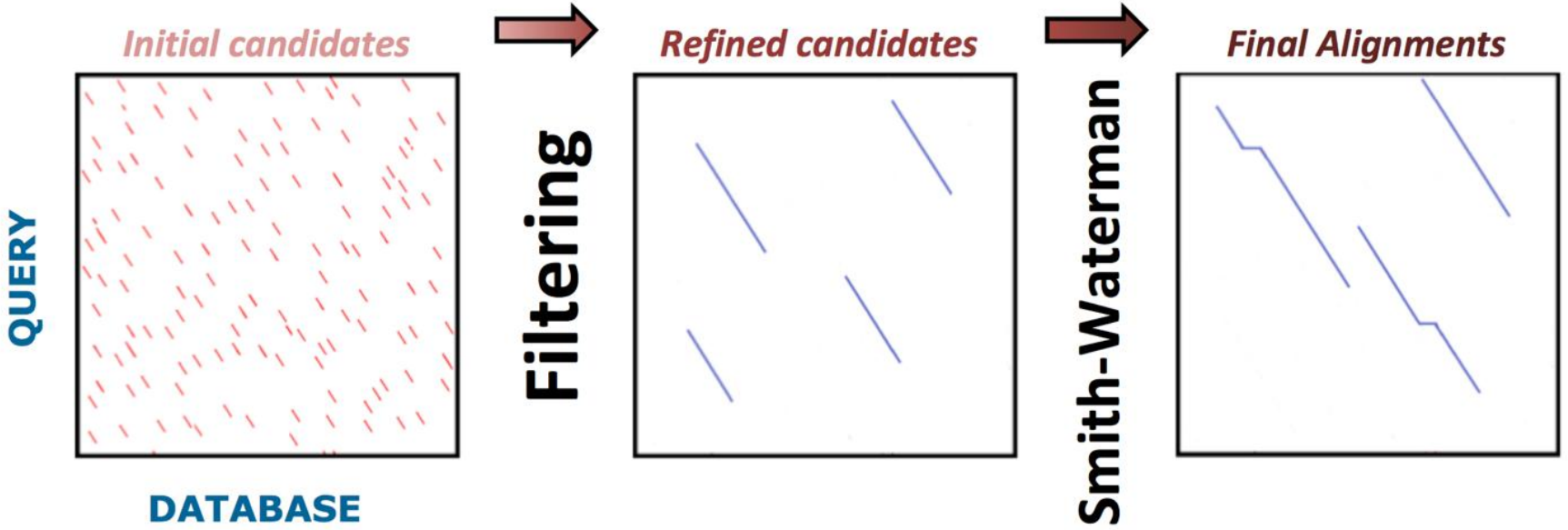
Local Alignment

Target Sequence
5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'
|||| |||||| ||||||||||||||||
Query Sequence 5' TACTCACGGATGAGGTACTTTAGAGGC 3'



SMITH - WATERMAN

BLAST (Basic local alignment search tool)



BLAST & Bioinformatics Databases

Searching millions of sequences in seconds to identify genes, proteins, and homologs

BLAST — Basic Local Alignment Search Tool

Published 1990 — NCBI (Altschul, Gish, Miller, Myers, Lipman)

The most cited paper in bioinformatics: >90,000 citations

The question BLAST answers:

"Does this sequence look like anything already known in the database?"

How it works — the heuristic shortcut:

Break query into short overlapping words (k-mers: 3 aa for proteins, 11 nt for DNA)

Find all similar k-mers above a threshold score (seeds)

Extend seeds in both directions while score increases (HSPs)

Apply gapped alignment to best High-Scoring Pairs

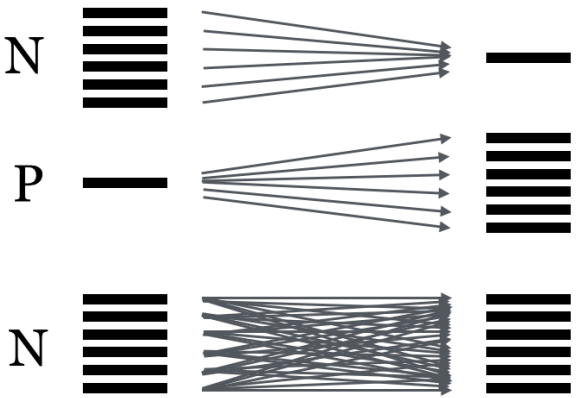
Calculate statistical significance (E-value) for each hit

Why not full Smith-Waterman?

Too slow for databases of 200 million sequences. BLAST heuristic is ~50× faster with minimal loss in sensitivity.

Comparisons of Sequences in Different Combinations

Program	Query Type	Subject Type	Computation
`blastn`	N	N	~ 1X
`blastp`	P	P	~ 1X
`blastx`	N	P	~ 6X
`tblastn`	P	N	~ 6X
`tblastx`	N	N	~36X



(other BLAST types not listed: `psiblast`, `deltablast`, `rpsblast`)

BLAST Variants — Choosing the Right Tool

Tool	Query	Database	Best for	Example use
BLASTp	Protein	Protein	Find similar proteins. Identify homologs across species.	Unknown protein from T cell biopsy → identify it
BLASTn	Nucleotide	Nucleotide	Find similar DNA/RNA sequences. Identify gene variants.	Patient tumour DNA → find mutations vs reference
BLASTx	Nucleotide	Protein	Translate DNA in all 6 frames, search protein DB.	New genomic region → discover which protein it encodes
tBLASTn	Protein	Nucleotide	Find DNA sequences that encode similar proteins.	Known protein → find gene in unsequenced genome
tBLASTx	Nucleotide	Nucleotide	Translate both query and DB, compare protein translations.	Compare coding regions across species at protein level

★ **For drug target work: BLASTp is your default. You have a protein and want to find homologs, conserved domains, and cross-species variants.**

Reading BLAST Output — What Each Number Means

★ E-value

The most important number

Expected number of hits this good by CHANCE in a database this size.

- E = 0.0 → impossible by chance → true match
- E < 1e-10 → very significant
- E < 0.05 → possibly significant
- E > 1 → likely noise, ignore

% Identity

Sequence similarity

Percentage of identical residues in the aligned region.

- 100% = perfect match
- 30-40% = likely same fold (homolog)
- <20% = twilight zone (careful)

Human vs Mouse CTLA-4: ~76% identity

% Coverage

How much is aligned

Percentage of YOUR query covered by the alignment.

- Low coverage (10%) + high identity = domain match only
- High coverage (>80%) = full-length homolog

Always check both identity AND coverage.

Bit Score

Normalised alignment score

Higher = better. Comparable across databases of different sizes. Used for ranking hits. A bit score >50 with low E-value = significant hit worth investigating.

Rule: E-value < 1e-5 AND Coverage > 80% AND Identity > 30% = likely true homolog

🔬 Exercise 1 — Identify an Unknown Protein Using BLASTp

💼 *Scenario: Your colleague found this sequence in a human T cell sample. What protein is it?*

Step 1: Go to blast.ncbi.nlm.nih.gov → click "Protein BLAST (BLASTp)"

Step 2: Paste the sequence below into the query box:

Step 3: Database: Non-redundant protein sequences (nr). Click BLAST.

Step 4: Look at the first hit: What protein? E-value? % Identity? Organism?

Step 5: Find the accession number — you will need it for Exercise 2.

https://blast.ncbi.nlm.nih.gov/Blast.cgi

Important update

The ***ClusteredNR*** database is now the **default** Protein BLAST database. **Learn more about ClusteredNR**

Basic Local Alignment Search Tool

BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. Learn more

NEWS

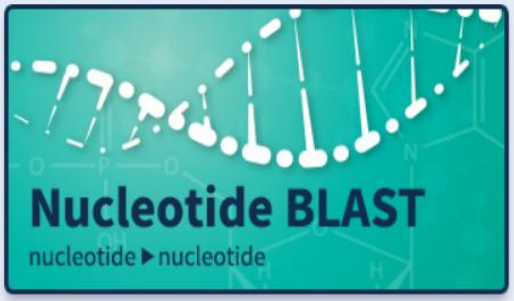
Mon, 21 Jul 2025

Here are a few highlights in our latest BLAST+ release:

Download BLAST+ 2.17.0 now!

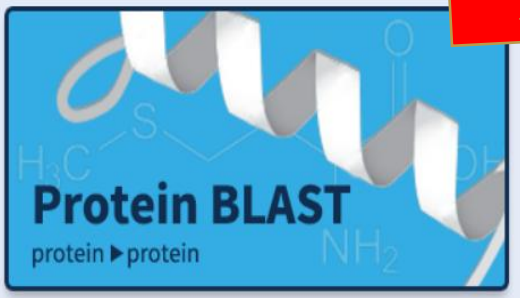
More BLAST news...

Web BLAST



blastx
translated nucleotide ▶ protein

tblastn
protein ▶ translated nucleotide



BLAST Genomes

Enter organism common name, scientific name, or tax id

Search

Files to use in this lesson



github.com/estevaobarcelos/From-Genome-to-Drug-Course

↓

Lesson_2_Mystery_Sequence_BLASTn

Lesson_2_Mystery_Sequence_BLASTp



What is this sequence and from which specie(s)?

- MEEPQSDPSVEPPLSQETFSDLWKLLPENNVLSPLPSQAMDDLMLSPDDIEQWFT
  EDPGPDEAPRMPEAAPPVAPAPAAPTPAAPAPAPSWPLSSSVPSQKTYQGSYGFR
  LGFLHSGTAKSVTCTYSPALNKMFCQLAKTCPVQLWVDSTPPPGTRVRAMAIYKQ
  SQHMTEVVRRCPHHERCSDSDGLAPPQHLIRVEGNLRVEYLDDRNTFRHSVVVPY
  EPPEVGSDCTTIHYNYMCNSSCMGGMNRRPILTIITLEDSSGNLLGRNSFEVRVC
  ACPGRDRRTEEENLRKKGEPHHELPPGSTKRALPNNTSSSPQPKKKPLDGEYFTL
  QIRGRERFEMFRELNEALELKDAQAGKEPGGSRAHSSHLKSKKGQSTSRHKKLMF
  KTEGPDSD

Lesson_2_Mystery_Sequence_BLASTp

National Library of Medicine
National Center for Biotechnology Information

estevaobarcelos@o...

BLAST ® » blastp suite

Home Recent Results Saved Strategies Help

Important update

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Standard Protein BLAST

blastn | **blastp** | blastx | tblastn | tb

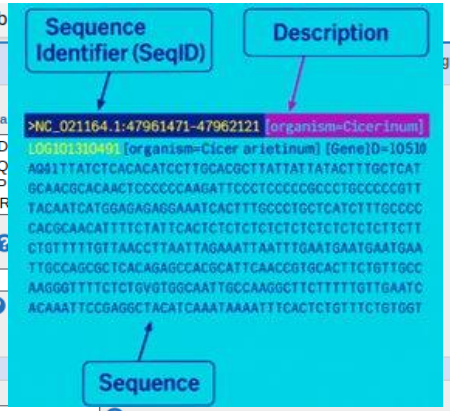
...grams search protein databases using a protein query. more...

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) Clea

MEEPQSDPSV...DDLMLSPDD
QWFTEDPGPD...VPLSSSVPSQ
YQGSYGFRLG...VQLWVDSTP
GTRVRAMAIY...QHLIRVEGNLR

FASTA format



Or, upload file Choose File No file chosen

Job Title

Enter a descriptive title for your BLAST search

Align two or more sequences

Choose Search Set

Database ClusteredNR (nr cluster seq)

Organism Optional

Enter organism name or id--completions will be suggested Add organism

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown.

Program Selection

Algorithm

- blastp (protein-protein BLAST)
- PSI-BLAST (Position-Specific Iterated BLAST)

Choose a BLAST algorithm

BLAST Search database ClusteredNR using Blastp (protein-protein BLAST)

Show results in a new window

+ Algorithm parameters

If you only want part of your sequence analyzed, enter start and end positions. Otherwise, leave blank.

Database	Description
clusteredNR	Compressed version of "non redundant protein sequences"
UniProtKB/Swiss-Prot	Manually curated, high-quality sequences
RefSeq select proteins	Curated representative proteins from NCBI
Protein Data Bank (pdb)	protein sequences from structures in the Protein Data Bank

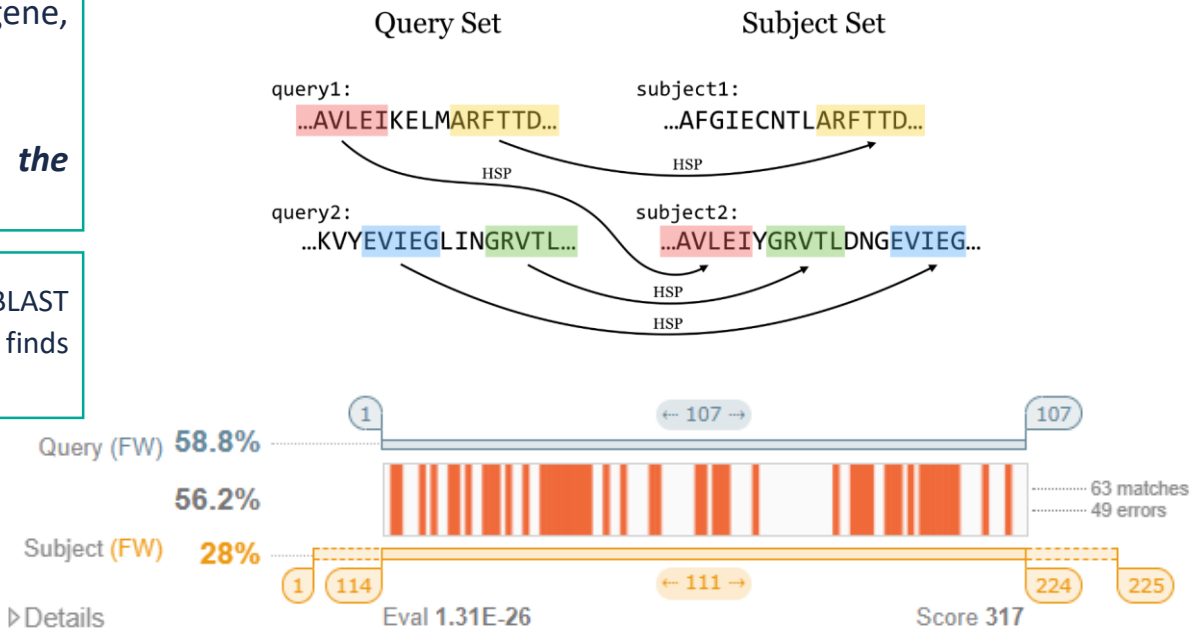
Algorith	Description
BLASTP	Finds **close and moderately distant homologs**
PSI-BLAST (Position-Specific Iterated BLAST)	Great for finding **remote homologs** that share low sequence identity (<25–30%)

Query vs Subject

QUERY: ***my sequence*** (e.g., gene, protein, fragment).

SUBJECT: ***sequence(s) from the database.***

Practical example: BLAST compares query vs. subject → finds ***homology***.



E-value: The BLAST E-value (Expectation value or Expect value) is a measure of the probability that your alignment arose just by chance. ***In practice, the E-value will always be extremely high (potentially in millions) for very short sequences, and very low (think 10^{-22} or even 0) for long sequences.***

Edit Search | Save Search | Search Summary | How to read this report? | BLAST Help Videos | Back to Traditional Results Page

Job Title	Protein Sequence
RID	GJ6M68MK016 *Search expires on 11-05 04:11 am* Download All
Program	BLASTP Citation
Database	ClusteredNR See details
Query ID	lcl|Query_9373571
Description	unnamed protein product
Molecule type	amino acid
Query Length	393
Other reports	Distance tree of results Multiple alignment MSA viewer

Filter Results

Organism *only top 20 will appear* NEW

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity [ ] to [ ]

E value [ ] to [ ]

Query Coverage [ ] to [ ]

Filter Reset

Clusters | Graphic Summary | Alignments | Taxonomy

Clusters producing significant alignments Download Select columns Show 100

select all *100 clusters selected* GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

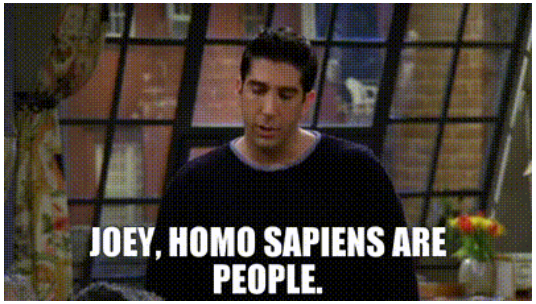
Cluster Composition (Click the ⊞ to see the cluster contents)	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
⊞ 99 member(s), 22 organism(s)	human	cellular tumor antigen p53 isoform a [Homo sapiens]	813	813	100%	0.0	100.00%	393	NP_000537.3
⊞ 1 member(s), 1 organism(s)	NA	cellular tumor antigen p53 isoform X1 [Macaca thibetana thibet...	785	785	100%	0.0	95.67%	490	XP_050621303.1
⊞ 9 member(s), 7 organism(s)	human	cellular tumor antigen p53 isoform g [Homo sapiens]	737	737	90%	0.0	100.00%	354	NP_001119590.1
⊞ 4 member(s), 4 organism(s)	NA	PREDICTED: cellular tumor antigen p53 isoform X2 [Colobus ...	724	724	100%	0.0	89.82%	369	XP_011813950.1
⊞ 15 member(s), 6 organism(s)	human	cellular tumor antigen p53 isoform c [Homo sapiens]	689	689	87%	0.0	97.65%	346	NP_001119585.1
⊞ 1 member(s), 1 organism(s)	green monkey	cellular tumor antigen p53 isoform X1 [Chlorocebus sabaeus]	689	689	94%	0.0	94.57%	406	XP_008008383.3
⊞ 1 member(s), 1 organism(s)	aye-aye	cellular tumor antigen p53 isoform a [Daubentonia madagasca...	687	687	100%	0.0	88.55%	389	KAL2765772.1
⊞ 1 member(s), 1 organism(s)	gray mouse lemur	cellular tumor antigen p53 [Microcebus murinus]	664	664	100%	0.0	86.01%	385	XP_012631512.1
⊞ 2 member(s), 2 organism(s)	Philippine flying lemur	cellular tumor antigen p53 [Cynocephalus volans]	664	664	100%	0.0	87.79%	391	XP_062967318.1
⊞ 9 member(s), 6 organism(s)	tufted capuchin	cellular tumor antigen p53 isoform X1 [Sapajus apella]	662	662	94%	0.0	90.54%	412	XP_032136321.1
⊞ 4 member(s), 3 organism(s)	Upper Galilee mountains ...	cellular tumor antigen p53 [Nannospalax galili]	661	661	100%	0.0	84.52%	391	NP_001306217.1
⊞ 4 member(s), 3 organism(s)	Bennett's brown lemur	cellular tumor antigen p53 isoform X2 [Eulemur rufifrons]	654	654	100%	0.0	82.95%	363	XP_069338125.1
⊞ 3 member(s), 2 organism(s)	rabbit	cellular tumor antigen p53 [Oryctolagus cuniculus]	651	651	100%	0.0	86.04%	391	NP_001075873.1

NCBI accession numbers — what the prefix tells you

Prefix	Sequence type	What it contains	CTLA4 example
NM_	**RefSeq mRNA** Mature transcript — no introns, with UTRs, curated manually	Most used for primer design, RT-PCR, gene expression. The "working sequence" of a gene	`NM_005214.5` CTLA4 mRNA (isoform 1)
NP_	**RefSeq protein** Amino acid sequence translated from the NM_ transcript	Always paired with NM_. NM_005214 → NP_005205 Use for protein BLAST (BLASTp)	`NP_005205.2` CTLA-4 protein 223 amino acids
NG_	**RefSeq genomic gene** Full gene locus — includes introns, exons, flanking regions	Used when you need intron sequences or full gene context. Much longer than NM_	`NG_011502.1` CTLA4 locus with all introns
NC_	**RefSeq chromosome** Complete chromosome sequence — billions of bases	CTLA4 is on chromosome 2. NC_000002 contains the full chr 2 — position 2q33.2	`NC_000002.12` Chromosome 2 entire sequence
NR_	**Non-coding RNA** lncRNA, miRNA, rRNA — RNAs that do not encode protein	Same curated system as NM_ but for functional non-coding RNA molecules	`NR_024540` 28S rRNA (not CTLA4)
XM_ XP_	**Predicted (computational)** Auto-generated by gene prediction algorithms, not curated	Less reliable than NM_/NP_. Use RefSeq (NM_, NP_) whenever available — avoid XM_/XP_	Found in poorly annotated genomes or novel species

Title	***What does this mean?***
Description	Sequence ***name*** found in the database (gene, organism, total/CDS)
Scientific Name	Scientific name of the organism
Max Score	Highest alignment score between query and database sequence (subject)
Total score	Sum of scores of all alignments with that sequence. Usually equal to the Max Score if there is only one alignment
Query Cover	Percentage of your sequence (query) that was ***covered*** by the alignment
E-value	Probability that the alignment occured by chance. ***The lower the probability, the more significant.*** Ex.; 0.0 = very high confidence
Per. Ident	Percentage of nucleotides in your sequnce that are ***identical*** to the sequence in the database
Acc. Len	Total length of the sequence in the database
Accession	Unique sequence identifier in NCBI, used for reference

Is this gene conserved in other species?



Homo sapiens
NP_000537.3 ***RefSeq Ptn***



Macaca thibetana thibetana



Chlorocebus sabaeus



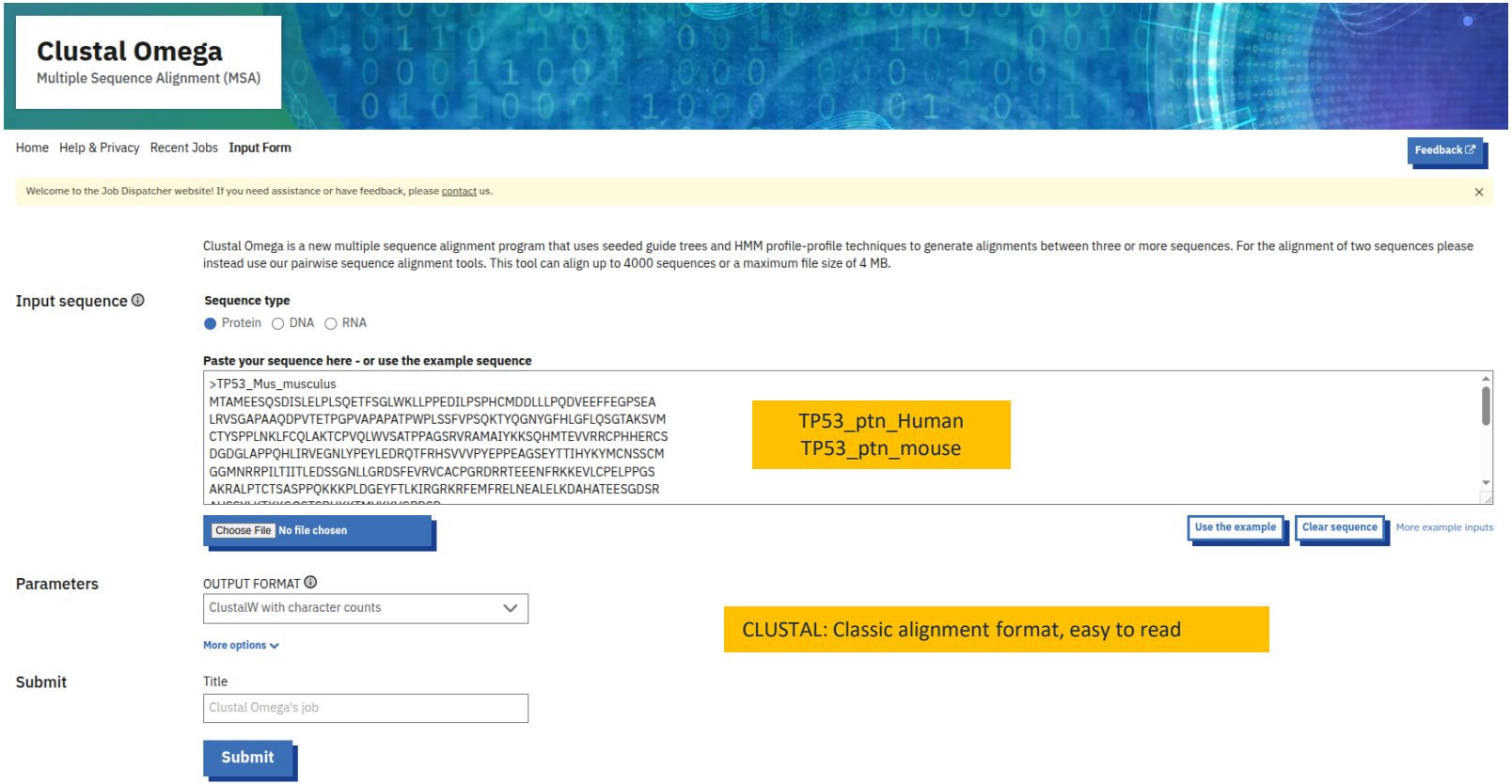
Microcebus murinus

Is the identity the same among all?

100% 95.6% 94.5% ...

Essential genes (those required for survival) are under strong purifying selection — harmful mutations are removed because they reduce fitness or cause lethality. These genes evolve slowly.

⍰ Exercise 2 — Alignment of sequences (Clustal Omega – Multiple Sequence Alignment)



Results of Protein Comparison

Alignment with colours

Hide

```
CLUSTAL O(1.2.4) multiple sequence alignment

TP53_Mus_musculus      MTAMEESQSDISLELPLSQETFSGLWKLLPPEDILPSP--HCMDD-LLLPQDVEEFFE--    55
TP53_Homo_sapiens      ---MEEPQSDPSVEPPLSQETFSDLWKLLPENNVLSPLPSQAMDDLMLSPDDIEQWFTED    57
                          *** *** *:* ********.****** :::*     :.*** :* *:*:*::*

TP53_Mus_musculus      -GPSEALRVSGAPAAQDPVTETPGPVAPAPATPWPLSSFVPSQKTYQGNYGFHLGFLQSG    114
TP53_Homo_sapiens      PGPDEAPRMPEAAPPVAPAPAAPTPAAPAPAPSWPLSSSVPSQKTYQGSYGFRLGFLHSG    117
                        **.** *:  *     *.  :* *.*****  ***** *********.***:****:**

TP53_Mus_musculus      TAKSVMCTYSPPLNKLFCQLAKTCPVQLWVSATPPAGSRVRAMAIYKKSQHMTEVVRRCP    174
TP53_Homo_sapiens      TAKSVTCTYSPALNKMFCQLAKTCPVQLWVDSTPPPGTRVRAMAIYKQSQHMTEVVRRCP    177
                       ***** ***** ***:**************.:*** *:*********:************

TP53_Mus_musculus      HHERCSDGDGLAPPQHLIRVEGNLYPEYLEDRQTFRHSVVVPYEPPEAGSEYTTIHYKYM    234
TP53_Homo_sapiens      HHERCSDSDGLAPPQHLIRVEGNLRVEYLDDRNTFRHSVVVPYEPPEVGSDCTTIHYNYM    237
                       *******.****************  ***:**:**************.**: *****:**

TP53_Mus_musculus      CNSSCMGGMNRRPILTIITLEDSSGNLLGRDSFEVRVCACPGRDRRTEEENFRKKEVLCP    294
TP53_Homo_sapiens      CNSSCMGGMNRRPILTIITLEDSSGNLLGRNSFEVRVCACPGRDRRTEEENLRKKGEPHH    297
                       ******************************:********************:***

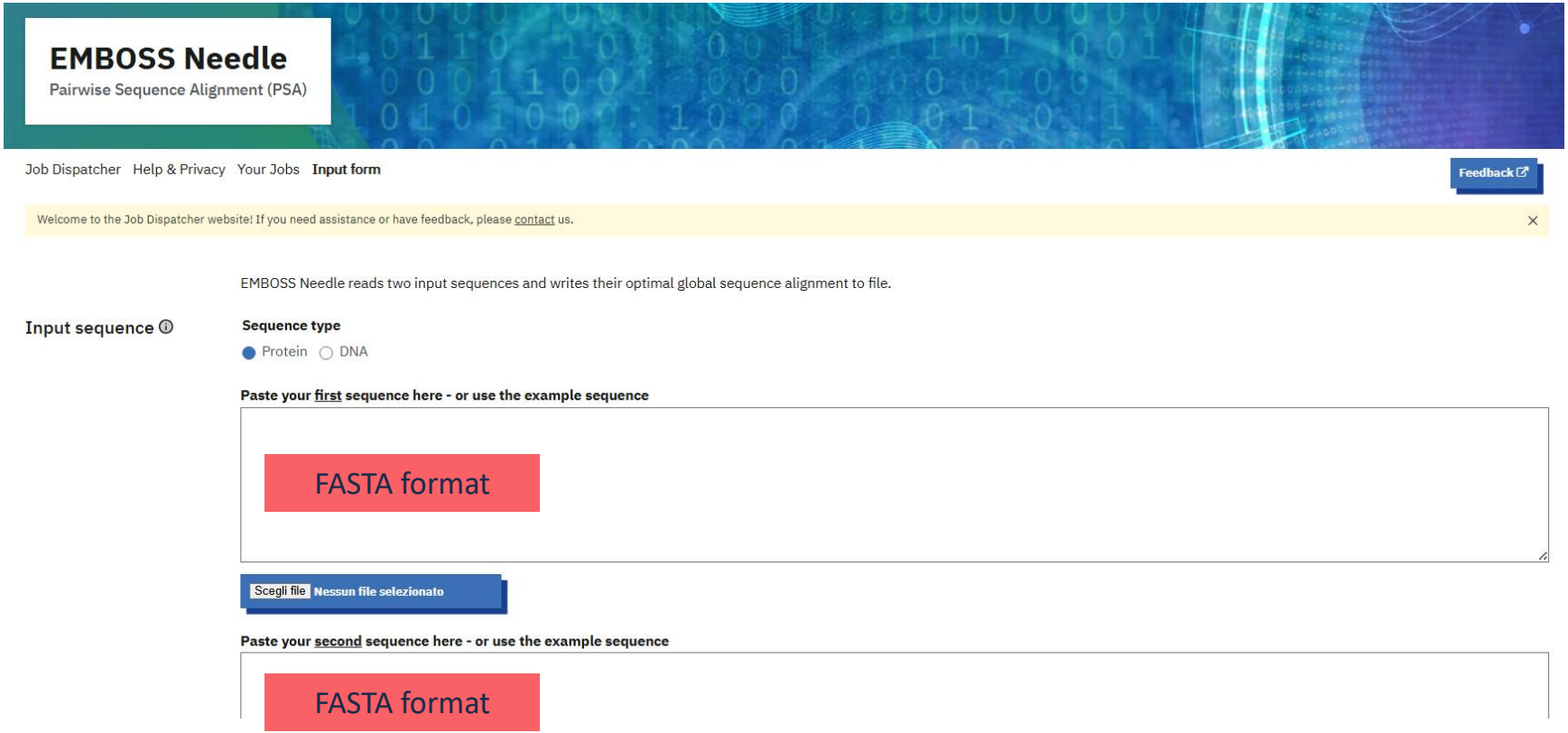
TP53_Mus_musculus      ELPPGSAKRALPTCTSASPPQKKKPLDGEYFTLKIRGRKRFEMFRELNEALELKDAHATE    354
TP53_Homo_sapiens      ELPPGSTKRALPNNTSSSPQPKKKPLDGEYFTLQIRGRERFEMFRELNEALELKDAQAGK    357
                       ******:*****. **:**  ************:****:******************:* :

TP53_Mus_musculus      ESGDSRAHSSYLKTKKGQSTSRHKKTMVKKVGPDSD     390
TP53_Homo_sapiens      EPGGSRAHSSHLKSKKGQSTSRHKKLMFKTEGPDSD     393
                       * *.******:**:*********** *.*. *****
```

- **Asterisk ***
  - Total identity: All aligned sequences have exactly the same amino acid at this position.
- **Colon :**
  - Conservative substitution: The amino acids are different, but have similar chemical properties (charge, polarity, hydrophobicity).
- **Period .**
  - Semi-conservative substitution: The amino acids are somewhat similar, but not completely equivalent.
- **Blank space**
  - Column with unrelated amino acids, or gaps.

⍰ Exercise 3 — Percent identity / Quantitative global alignment

EMBOSS Needle (Pairwise Sequence Alignment)



Results of Global Sequence Analysis

Tool Output | Result Files

Tool output

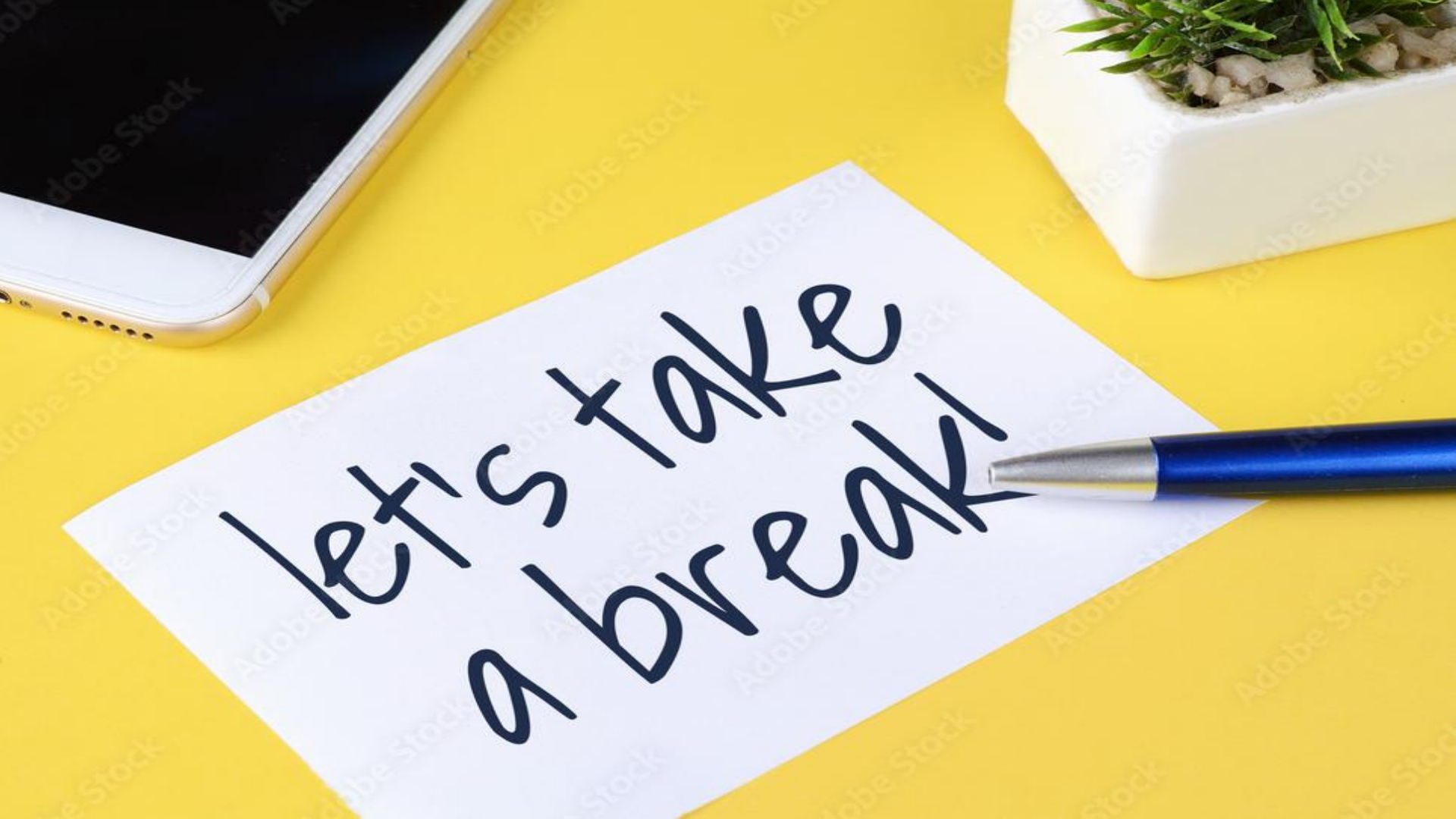
Download

```
########################################
# Program: needle
# Rundate: Tue  4 Nov 2025 16:24:29
# Commandline: needle
#    -auto
#    -stdout
#    -asequence emboss_needle-I20251104-162421-0379-22388156-p1m.asequence
#    -bsequence emboss_needle-I20251104-162421-0379-22388156-p1m.bsequence
#    -datafile EBLOSUM62
#    -gapopen 10.0
#    -gapextend 0.5
#    -endopen 10.0
#    -endextend 0.5
#    -aformat3 pair
#    -sprotein1
#    -sprotein2
# Align_format: pair
# Report_file: stdout
########################################

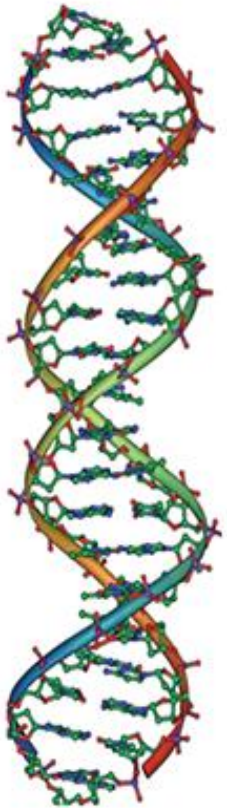
#=======================================
#
# Aligned_sequences: 2
# 1: TP53_Mus_musculus
# 2: TP53_Homo_sapiens
# Matrix: EBLOSUM62
# Gap_penalty: 10.0
# Extend_penalty: 0.5
#
# Length: 396
# Identity:     304/396 (76.8%)
# Similarity:   326/396 (82.3%)
# Gaps:           9/396 ( 2.3%)
# Score: 1558.0
#
#
#=======================================
```

- **Length Total alignment length: 396 amino acids**
- **Identity Identical residues (same in both sequences) 304 / 396 → 76.8%**
- **Similarity Similar residues (same chemical type) 326 / 396 → 82.3%**
- **Gaps Gaps introduced to optimize the alignment 9 / 396 → 2.3%**
- **Score Total alignment score (higher = greater similarity) 1558.0.**

In practice, the two proteins share approximately 77% identity and 82% overall similarity, with very few gaps: this means that the sequences (and therefore the biological functions) are highly evolutionarily conserved.

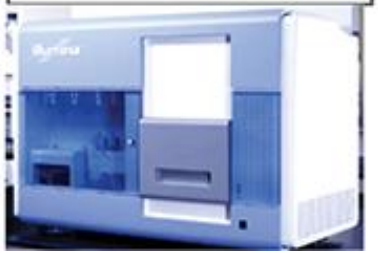

let's take
a break!

Human genome



Short reads

.....
GGTCTGGATGC
CGGTCTGGATGC
GCGGTCTGGATG
GCGGTCTGGAT
GGCGGTCTGGAT
GGCGGTCTGGA
TCTATGCGGGCCCCT
TCTATGCGGGCCCC
ATCTATGCGGGCC
TATCTATGCGGGC
TTATCTATGCGGG
CTTATCTATGCGGG



Alignment of reads to the reference genome and SNP calling

SNP:A->G

GTCTGGATGCT TCTATGCGGGCCCCT
GGTCTGGATGC TCTATGCGGGCCCC
CGGTCTGGATGC ATCTATGCGGGCC
GCGGTCTGGATG TATCTATGCGGGC
GCGGTCTGGAT TTATCTATGCGGG
GCGGTCTGGAT CTTATCTATGCGGG
GGCGGTCTGGAT CTTATCTATGCGG
GGCGGTCTGGA CTTATCTATGCGG

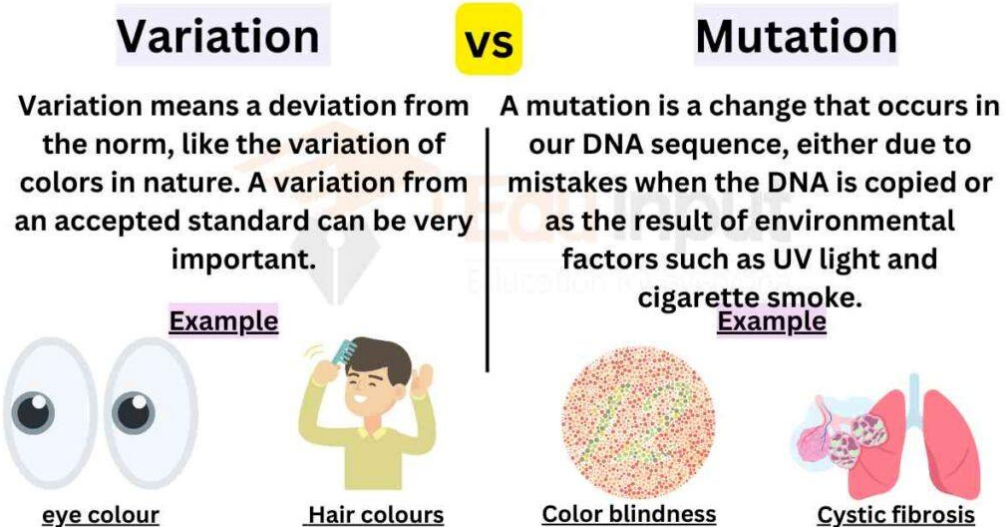
GGCGGTCTAGATGCTTATCTATGCGGGCCCCT

Reference genome sequence

Mutations & Polymorphisms

From a single nucleotide change to disease, drug resistance, and drug targets

Polymorphisms and Mutations



Reference SNP
rs1111111

Types of Mutations — What Can Go Wrong?

Silent

Base change, same amino acid. No protein change — genetic code is redundant.
GAA → GAG = Glu → Glu ✓

Missense

Base change → different amino acid. Impact depends on position.
GAA → GTA = Glu → Val (charged → hydrophobic)

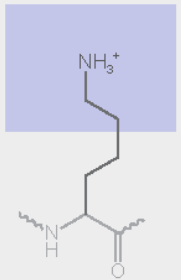
Nonsense

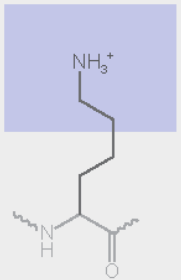
Base change creates STOP codon. Protein truncated — usually non-functional.
CAA → UAA = Gln → STOP

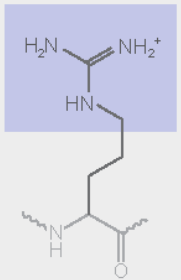
Frameshift

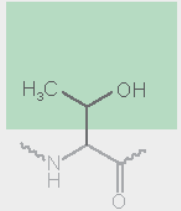
Insertion/deletion not divisible by 3. Shifts reading frame — all downstream codons wrong.

	No mutation	Point mutations			
		Silent	Nonsense	Missense	
				conservative	non-conservative
DNA level	TTC	TTT	ATC	TCC	TGC
mRNA level	AAG	AAA	UAG	AGG	ACG
protein level	**Lys**	**Lys**	**STOP**	**Arg**	**Thr**

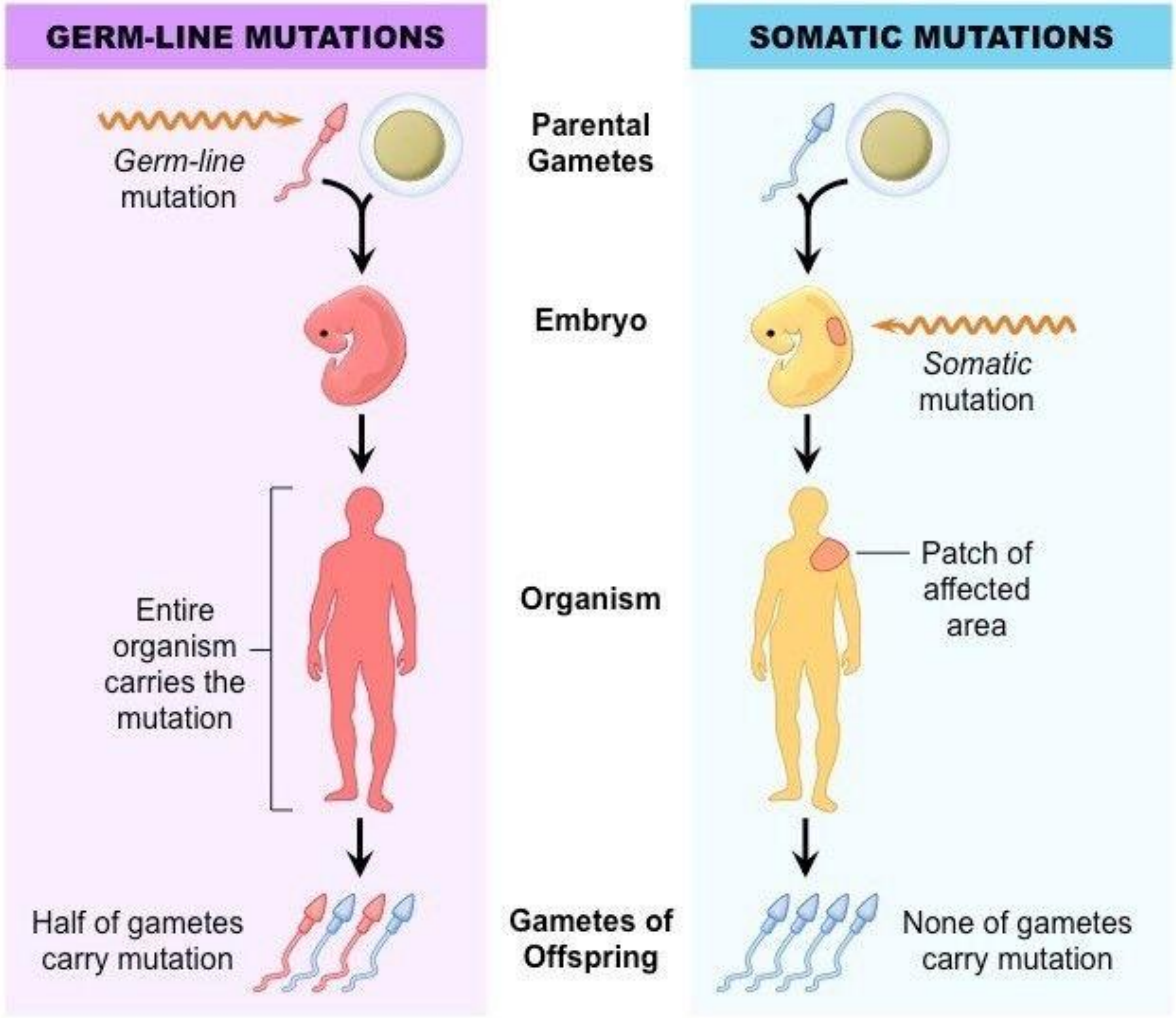








basic
polar


GERM-LINE MUTATIONS
SOMATIC MUTATIONS
Parental Gametes
Germ-line mutation
Embryo
Somatic mutation
Entire organism carries the mutation
Organism
Patch of affected area
Half of gametes carry mutation
Gametes of Offspring
None of gametes carry mutation

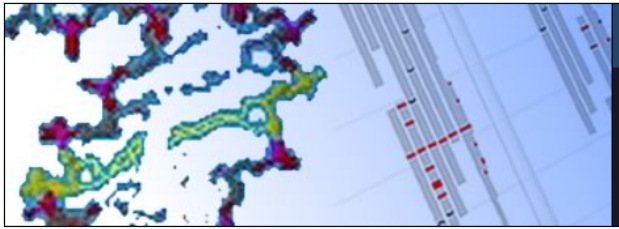
dbSNP (Single Nucleotide Polymorphism Database)

An official website of the United States government Here's how you know

National Library of Medicine
National Center for Biotechnology Information

Log in

dbSNP SNP Search
Advanced Help



dbSNP

dbSNP contains human single nucleotide variations, microsatellites, and small-scale insertions and deletions along with publication, population frequency, molecular consequence, and genomic and RefSeq mapping information for both common variations and clinical mutations.

Getting Started

- dbSNP 25th Anniversary
- Overview of dbSNP
- About Reference SNP (rs)
- Factsheet
- FAQ
- Entrez Updates (May 26, 2020)

Submission

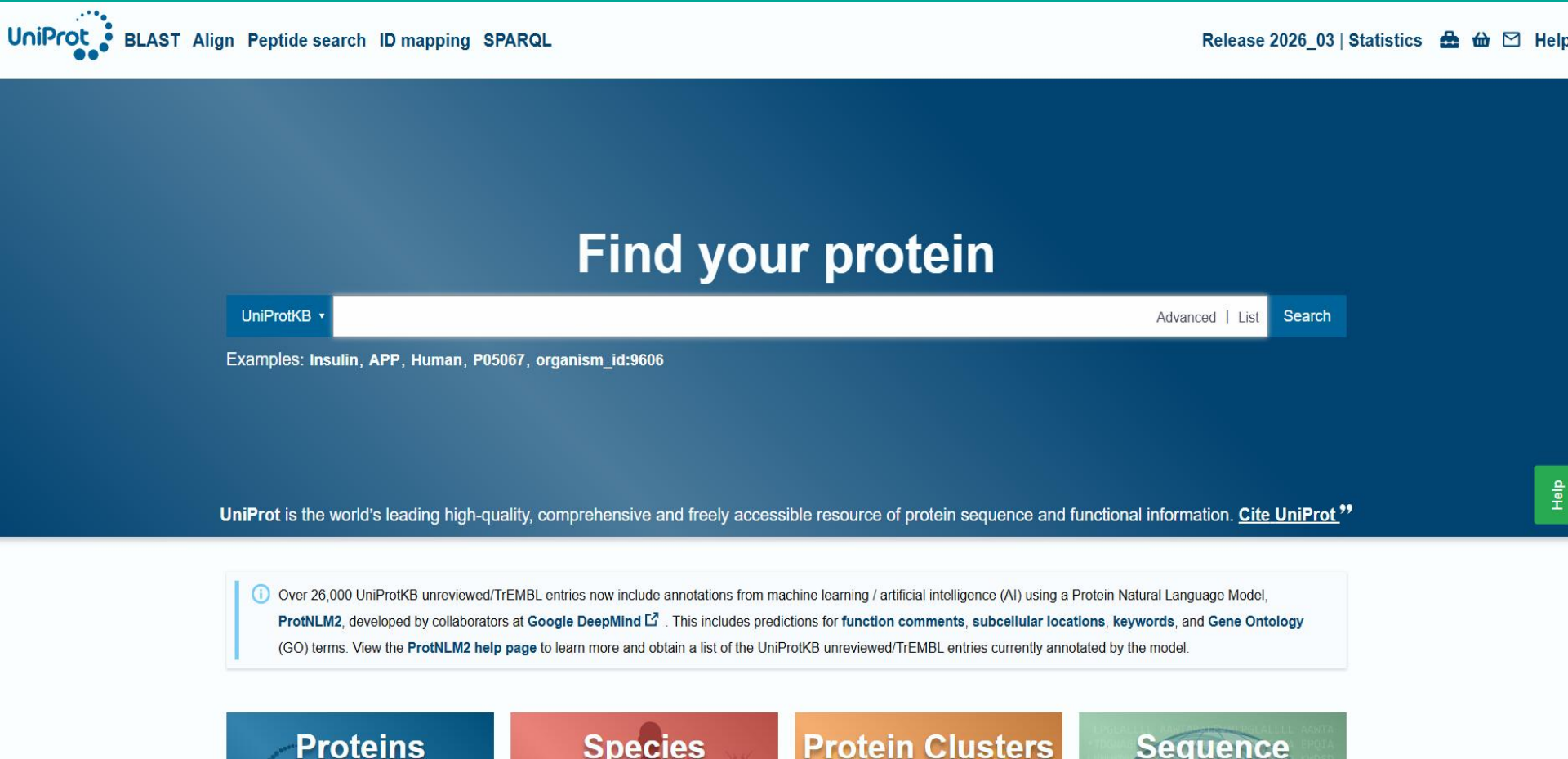
- How to Submit
- Hold Until Published (HUP) Policies
- Submission Search

Access Data

- Web Search
- eUtils API
- Variation Services
- FTP Download
- Tutorials on GitHub

rs606231422

UniProt (Universal Protein Resource)



R Practical — Sequence Alignment

Performing alignment, mutation analysis, in R